

I recommend cleaning it in **Power Query before creating the Power BI visuals** table contains fields such as Trip_ID, Vehicle_ID, Driver_ID, Origin, Destination, Distance_km, Fuel_Consumed_L, Delivery_Status, and Delivery_Date.

1. Open Power Query

In Power BI Desktop:

Home → Transform data → Transform data

Select your logistics/trips table.

2. Remove completely blank rows

Your source contains blank rows after the trip records, so these should be removed.

In Power Query:

Home → Remove Rows → Remove Blank Rows

This prevents empty records from entering your data model.

3. Check the data types

Set each column to the appropriate type:

Column	Recommended type
Trip_ID	Text
Vehicle_ID	Text
Driver_ID	Text
Origin	Text
Destination	Text
Distance_km	Whole Number
Fuel_Consumed_L	Decimal Number
Delivery_Status	Text
Delivery_Date	Date

4. Clean the text columns

Select:

Trip_ID, Vehicle_ID, Driver_ID, Origin, Destination, Delivery_Status

Then use:

Transform → Format → Trim

After that:

Transform → Format → Clean

Trim removes unwanted spaces before/after values, while Clean removes non-printing characters.

5. Handle missing Fuel Consumed values

This is particularly important in your dataset. For example, T016 has a blank Fuel_Consumed_L value.

I **would not replace these blanks with 0** unless zero genuinely means the vehicle consumed no fuel. Doing so could distort fuel-efficiency calculations.

You can leave the value as null and create your measure with:

Total Fuel Consumed =

SUM(Trips[Fuel_Consumed_L])

Power BI's SUM will ignore blank values.

If the assignment specifically requires missing-value treatment, you could instead filter out records with missing fuel **only for fuel-efficiency analysis**, while retaining them for trip-count analysis.

6. Check for duplicate Trip IDs

Select:

Trip_ID

Then use:

Home → Keep Rows → Keep Duplicates

This lets you inspect duplicates first.

If Trip_ID is supposed to uniquely identify each trip and the duplicate rows are accidental, remove them using:

Home → Remove Rows → Remove Duplicates

Don't remove duplicates automatically without checking why they exist.

7. Standardize Delivery Status

Your data uses values such as On-Time and Late.

Check the filter dropdown on Delivery_Status for variations such as:

On-Time

On Time

on-time

Late

late

Standardize them to:

On-Time

Late

You can use **Transform → Replace Values**.

8. Validate Distance and Fuel values

Filter Distance_km and Fuel_Consumed_L and look for:

- Negative numbers
- Zero values
- Null values
- Extremely unusual values
- Conversion errors

For example, your source contains legitimate-looking distances ranging from short trips such as 74 km to much longer trips such as 1,729 km.

Don't automatically delete large values merely because they look like outliers; verify whether they are valid trips first.

9. Check Origin and Destination

Check for inconsistent city names such as:

Delhi

delhi

DELHI

Delhi

A simple approach is:

Select Origin and Destination separately → **Transform** → **Format** → **Capitalize Each Word**.

That gives consistent values such as:

Delhi

Mumbai

Bangalore

Hyderabad

Chennai

Kolkata

Pune

10. Add useful date columns

After cleaning Delivery_Date, you can add:

Add Column → **Date** → **Year** → **Year**

Then:

Add Column → **Date** → **Month** → **Name of Month**

You can also create:

- Year
- Month
- Month Number
- Day

- Quarter

These will help with your delivery trend visualizations.

11. Finish the cleaning

When everything looks correct:

Home → Close & Apply

A good final Power Query table should have **no completely blank rows, correct data types, standardized city/status text, checked duplicates, valid dates, and properly handled missing fuel values.**

Create the specified visualizations in Power BI

Your uploaded dataset is a **transport/logistics trip dataset** containing fields such as Trip_ID, Vehicle_ID, Driver_ID, Origin, Destination, Distance_km, Fuel_Consumed_L, Delivery_Status, and Delivery_Date.

You can build a useful **Power BI Logistics Dashboard** from it as follows.

Visualization	Power BI fields	Purpose
Card – Total Trips	Trip_ID → Count	Number of trips
Card – Total Distance	Distance_km → Sum	Total distance travelled
Card – Total Fuel	Fuel_Consumed_L → Sum	Total fuel consumed
Donut Chart	Legend: Delivery_Status; Values: Count of Trip_ID	On-Time vs Late deliveries
Clustered Column Chart	X-axis: Vehicle_ID; Y-axis: Sum of Distance_km	Distance travelled by vehicle
Bar Chart	Y-axis: Driver_ID; X-axis: Count of Trip_ID	Trips completed by driver
Line Chart	X-axis: Delivery_Date; Y-axis: Count of Trip_ID	Trip activity over time
Column Chart	X-axis: Origin; Y-axis: Sum of Fuel_Consumed_L	Fuel consumption by origin
Matrix	Rows: Origin; Columns: Destination; Values: Count of Trip_ID	Route analysis
Slicers	Vehicle_ID, Driver_ID, Delivery_Status, Delivery_Date	Dashboard filtering

1. Create the measures first

Go to **Modeling** → **New measure** and create:

Total Trips =

DISTINCTCOUNT(Trips[Trip_ID])

Total Distance =

SUM(Trips[Distance_km])

Total Fuel Consumed =

SUM(Trips[Fuel_Consumed_L])

For fuel efficiency:

Fuel Efficiency =

DIVIDE(

[Total Distance],

[Total Fuel Consumed],

0

)

This produces **km per litre**.

For late deliveries:

Late Deliveries =

CALCULATE(

[Total Trips],

Trips[Delivery_Status] = "Late"

)

For on-time deliveries:

On-Time Deliveries =

CALCULATE(

[Total Trips],

Trips[Delivery_Status] = "On-Time"

)

If your table is not called Trips, replace Trips with your actual Power BI table name.

2. Create the Delivery Status Donut Chart

Select **Donut chart** from Visualizations.

Set:

Legend

Delivery_Status

Values

Trip_ID → Count

Title:

Delivery Status – On-Time vs Late

Your data contains both statuses—for example, T005 is On-Time while T006 is Late.

3. Distance by Vehicle — Clustered Column Chart

Select **Clustered column chart**.

X-axis

Vehicle_ID

Y-axis

Total Distance

Title:

Total Distance Travelled by Vehicle

This lets you compare vehicles such as V01, V02, V04, V05, V06 and V07 that appear in the dataset.

4. Trips by Driver — Bar Chart

Select **Clustered bar chart**.

Y-axis

Driver_ID

X-axis

Total Trips

Then select the visual → ... → **Sort by Total Trips → Descending**.

Title:

Number of Trips by Driver

5. Delivery Trend — Line Chart

Select **Line chart**.

X-axis

Delivery_Date

Y-axis

Total Trips

Make sure Delivery_Date has the **Date** data type rather than Text.

Title:

Delivery Trips Over Time

6. Origin-to-Destination Matrix

Select **Matrix**.

Rows

Origin

Columns

Destination

Values

Total Trips

This gives you a route matrix such as Delhi → Pune, Mumbai → Bangalore, Mumbai → Pune, Hyderabad → Pune, etc., which are present in your source data.

7. Add slicers

Create four separate **Slicer** visuals:

Vehicle_ID

Driver_ID

Delivery_Status

Delivery_Date